

Meeting of the Technical Steering Committee (TSC) Board

Tuesday, July 17th 2018
11:00am ET

Meeting Logistics

- https://zoom.us/j/556149142

- United States : +1 (646) 558-8656
  - Meeting ID: 556 149 142

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Agenda

- TSC 2018-2019:
  - reminder for current TSC members: please send nominations for Project Lead to Jeff by end of day tomorrow: Wednesday July 18, 2018

- Follow up from previous meetings:
  - ✓ SC'18 BoF submission: initial submission completed; sent link out to TSC with abstract/description text. We have until July 31 to amend
  - ✓ Hyperion research study: feedback from last time provided to Hyperion…several questions targeting respondents who have used OpenHPC will be added

- Quick update on some .spec file cleanup

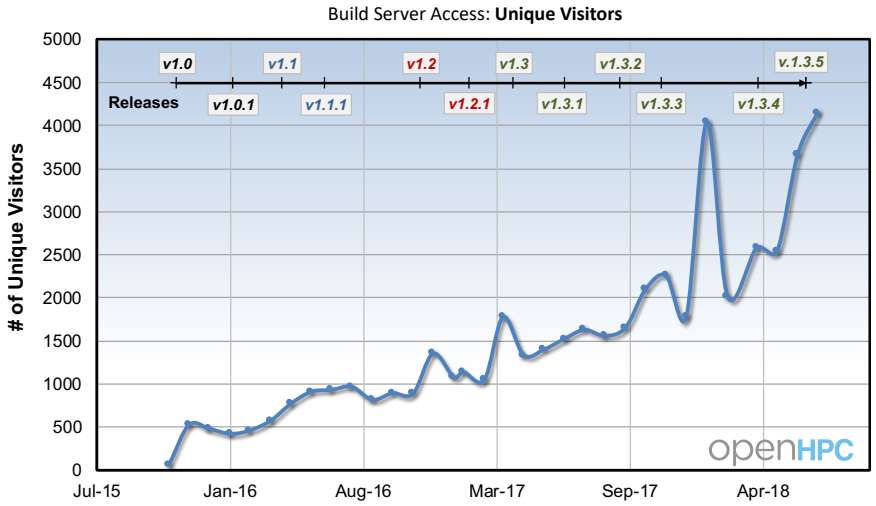
- Updated access stats

- Discussion on modulefile conventions

Quick update on some .spec file changes

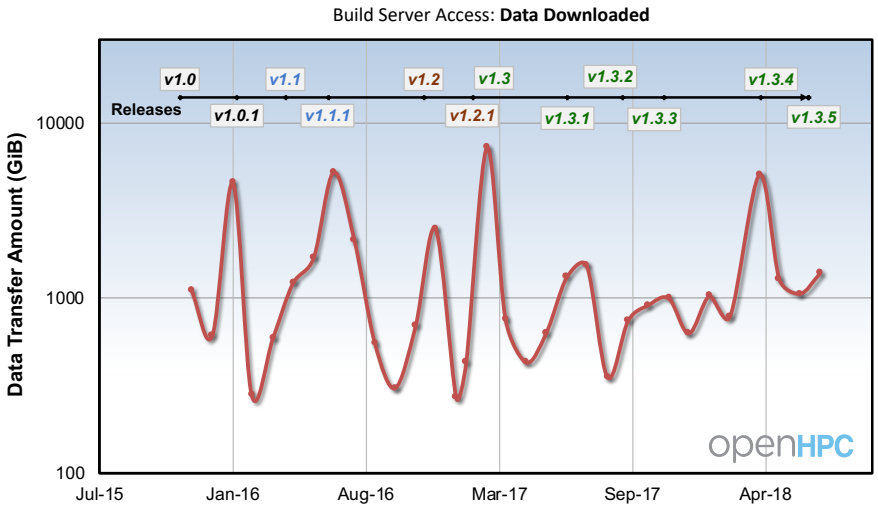
- A number of small changes were made to cleanup and update .spec files on the v1.3.6 branch (thanks Adrian)
  - removed BuildRoot definition (not needed)
  - removed duplicate definition of DocDir (it comes from OHPC_macros)
  - remove extra removal of %{buildroot} in %install section
  - remove %clean section (it's automatic)
  - removed need to define `Source[1-9]: OHPC_macros` in every .spec file that uses the macros
    - now defined within OHPC_macros directly which we already %include in necessary .spec files
    - do have to pick an index for the source designation, chose a larger number since we often use Source1, Source2, Source3, etc…
- new Travis check is now enabled on PRs to ensure that there is no conflict; also checks other .spec file conventions per the cleanup
  - ensure %changelog is not present
  - ensure DocDir is not redefined
  - ensure Buildroot not defined
  - etc.

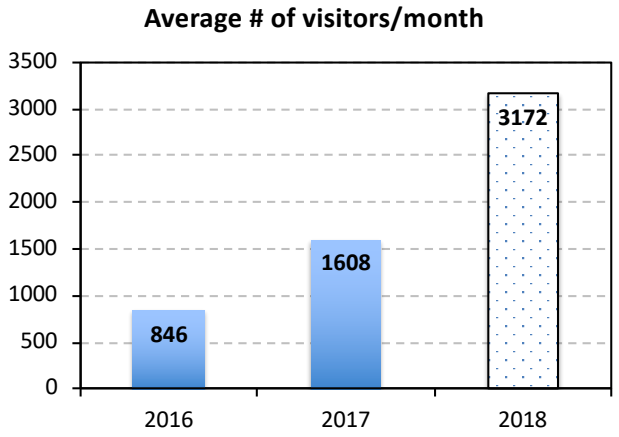
Updated Usage/Access Statistics (thru June 2018)



- Stats for build/repo server (tracking # of unique visitors per month and amount of data downloaded):

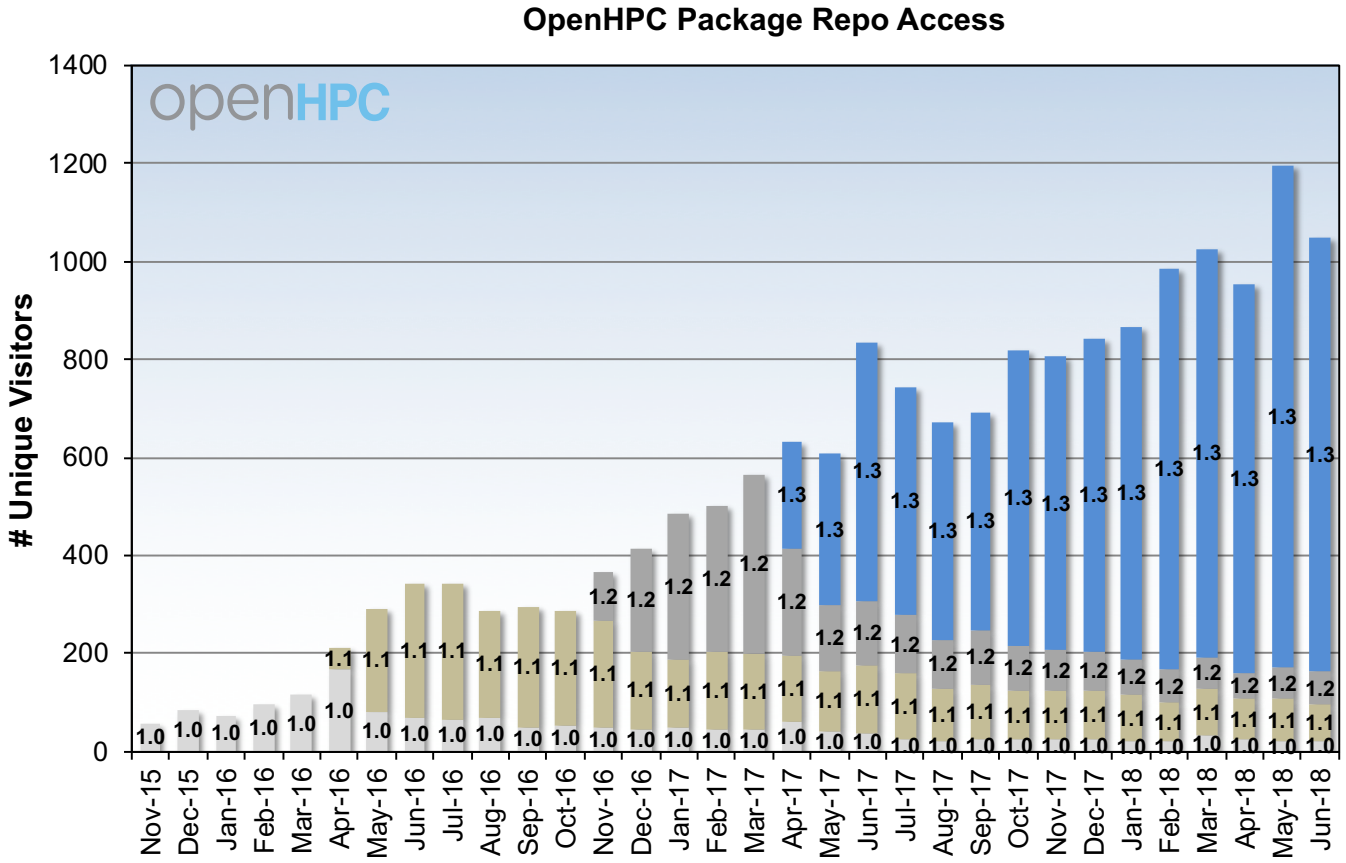
  http://build.openhpc.community





Updated Usage/Access Stats (thru June 2018)

- These stats monitor access specifically to package repository metadata (typically expected to be via yum/zypper)
- Repo access binned by minor version



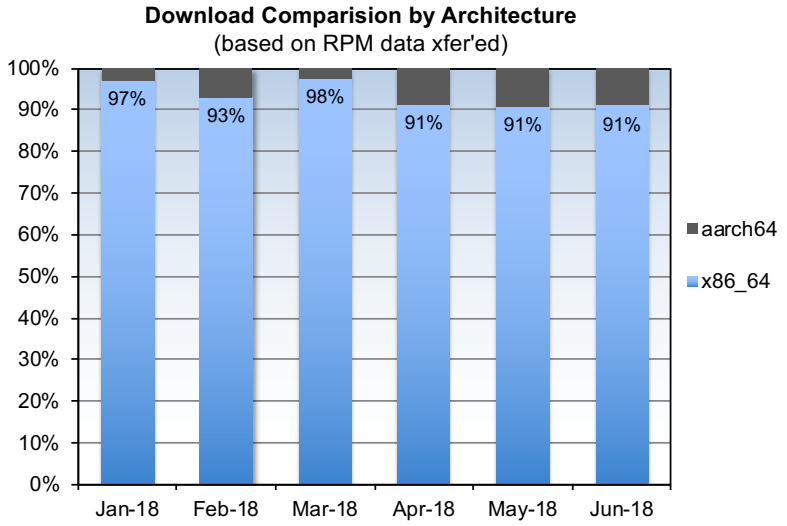
Updated Usage/Access Stats (thru June 2018)

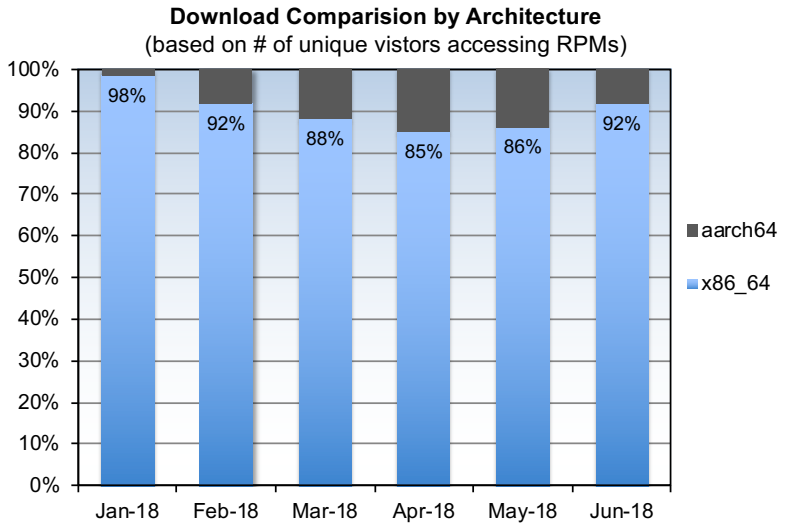
Architecture specific metrics:

- To provide some characterization, we scrape the access logs to analyze two architecture specific file types:

  ```
  (aarch64|x86_64).rpm
  (aarch64|x86_64).tar
  ```

- Plots compare percentages for the amount of data xfer'ed and the # of unique visitors accessing the (aarch64|x86_64) files





Modulefile conventions

- A question spurred by PR has come up on GitHub regarding conventions for exposing header files for compilation (https://github.com/openhpc/ohpc/pull/791)
- Recall that we have endeavored to provide a consistent set of environment variables for use by end users to highlight the top level installation path, path to provided binaries, path to provide libraries, and path to development headers, respectively.

  As an example for HDF5:

```
HDF5_DIR -> /opt/ohpc/pub/libs/gnu7/hdf5/1.10.2
HDF5_BIN -> /opt/ohpc/pub/libs/gnu7/hdf5/1.10.2/bin
HDF5_LIB -> /opt/ohpc/pub/libs/gnu7/hdf5/1.10.2/lib
HDF5_INC -> /opt/ohpc/pub/libs/gnu7/hdf5/1.10.2/include
```

- We generally check for existence and correctness of these vars in the test suite; intent is for users to be able to leverage these vars in their own build system or on command line and we do that as well in test suite.

  $ gcc superfly_test.c –I${HDF5_INC} –L${HDF5_LIB} –lhdf5

- For packages with binaries and dynamic libraries, we also augment standard search paths like:
  - $PATH
  - $LD_LIBRARY_PATH

- We also tend to set INCLUDE with the path to library header files and this is the primary focus of the PR commentary
  - compilers don't use this variable for anything
  - instead, gcc and llvm leverage the following vars:
    - CPATH: Paths given via this variable apply to all supported languages and are appended at the end after all paths given via -I.
    - C_INCLUDE_PATH: Paths given via this variable apply to C only and are appended at the end after all paths given via -isystem, which are processed after all paths given via -I.
    - CPLUS_INCLUDE_PATH: This variable is the same as above, but for C++.

Modulefile conventions

- So, we provide INCLUDE (as a duplicate of FOO_INC) but it doesn't do much
- PR is focused on amending to define CPATH instead
  - in theory, this means an end user could load module X and do compilation without leveraging any module-provided environment vars
  - Unknowns:
    - do ARM HPC and Intel compilers honor CPATH in the same way?
    - what about search paths for F90 modules?

- Thoughts?

- Potential Options:
  - do nothing - yes we currently set an extra variable, but it's not really hurting anything (although we don't do any testing with it)
  - remove setting of INCLUDE and continue to rely on $FOO_INC (which we test and advertise to users)
  - set CPATH in addition to our existing env vars (testing here could be a bit odd)
    - assuming we want to test both methods, we'd have to unset CPATH to test env variables
    - also means significant lift to the test suite to test both approaches (and runtime)
  - set CPATH instead of environment variable for development headers and rely on it exclusively (by users and for testing)
    - this is a big convention change for existing users
    - slight addendum is that we still provide the env vars for convenience (but not test)